

An exact mean bound for the rational complexity of binary strings

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Abstract

For a residue r modulo 2^N , let $R_N(r)$ be the least height $\max(|a|, b)$ of a rational representation $a \equiv rb \pmod{2^N}$ with positive odd denominator b . We prove that the uniform mean of the ordinary, unrounded quantity $\log_2 R_N(r)$ is at most $N/2$ for every $N \geq 1$. This is the mean part of Question 3.1 in the 2026 survey of Lin, Xiao and Chen. The proof compares the product of the two lift heights with a truncated logarithmic excess at the parent modulus. Short lattice vectors with even denominator cover the large-height residues; completion in the actual odd coset bounds their contribution. A finite lattice sum, a two-prime sieve and an explicit primitive-point estimate close the inequality for parent length at least 26. Integer certificates cover the remaining parent lengths $1, \dots, 25$. The variance question is left open.

1 The problem and the result

Write a binary string as $s = (s_0, \dots, s_{N-1})$, and identify it with $r = \sum_{i=0}^{N-1} s_i 2^i$ modulo 2^N . Define

$$R_N(r) = \min_{\substack{a \equiv rb \pmod{2^N} \\ b > 0, b \text{ odd}}} \max(|a|, b), \quad \mathcal{E}_N = 2^{-N} \sum_{r=0}^{2^N-1} \log_2 R_N(r). \quad (1.1)$$

This is the rational height underlying ordinary finite 2-adic complexity. The logarithm is not rounded to an integer. The measure includes every string, including the zero string; it is not a measure on odd residues.

Theorem 1.1. *For every integer $N \geq 1$,*

$$\mathcal{E}_N \leq \frac{N}{2}.$$

Tian and Qi studied this same mean and proved a weaker upper bound [1, Theorem 3]. Chen and Winterhof proved the lower bound $\mathcal{E}_N \geq N/2 - 1$ and an upper bound with a logarithmic error [2, Theorems 2–3]. Their short primitive counts credit Tian–Qi; their growth recurrence credits Goresky–Klapper [6]. The exact upper in Theorem 1.1 is explicitly asked for in [4, Question 3.1], together with a separate variance problem. We answer only the mean part.

Our proof uses elementary geometry of congruence lattices and finite integer certificates. It does not use a limiting distribution to infer an inequality at every length. Its computational premises are stated in Section 6 and checked by the accompanying dependency-free program. The uniform argument is a proof for all sufficiently large lengths, not an extrapolation of those certificates.

Source and priority boundary. For Chen–Winterhof’s detailed statements we use the explicitly identified arXiv version 1 [2]. The final publication [3] has been identified, but its full body has not been obtained for this paper. We do not assert a lemma-by-lemma identity of those versions. The July 2026 survey, which includes Chen as an author, explicitly poses the exact mean question after the final publication. No exact antecedent has been identified in the primary sources checked for this paper; this is not a certification of universal priority. Ren–Winterhof’s recent symmetric complexity [5] minimizes also over a reversed string and is a different random variable.

Notation. All logarithms in the proof are natural. The subscript in R_N denotes a length. When a parent modulus is $M = 2^n$, abbreviate $R_M(r) = R_n(r)$; thus R_{2M} denotes the child height R_{n+1} . Put

$$q = \sqrt{M}, \quad s = q/\sqrt{2}, \quad J = \lfloor n/2 \rfloor.$$

The convention $[x]_+ = \max(x, 0)$ will be used. A minimum in (1.1) is primitive in $\mathbb{Z}^2$: its common divisor is odd and invertible modulo 2^N , so division would otherwise lower its height. In particular $R_M(0) = 1$ and $R_M(r) \leq M/2$ for $M \geq 2$.

2 The two lifts and a logarithmic budget

Lemma 2.1. *Let $M = 2^n$, $n \geq 1$. For a residue r modulo M , put*

$$h = R_M(r), \quad H = \max\{R_{2M}(r), R_{2M}(r + M)\}.$$

Then $\min\{R_{2M}(r), R_{2M}(r + M)\} = h$, and

$$hH \leq \max(M, 2h^2). \quad (2.1)$$

Proof. The union of the two sets of child representations is exactly the set of parent representations, since an odd denominator is invertible modulo $2M$. Choose a primitive parent minimum $u = (a, b)$, $b > 0$ odd. It selects a child r_0 with height h . Reflection allows $a \geq 0$.

Suppose first $a > 0$. Integer solutions $v = (c, d)$ of $ad - bc = M$ form $v_0 + ku$ by Bézout's identity. The restriction d odd selects exactly one class of k modulo 2. Moreover

$$c \equiv (r_0 + M)d \pmod{2M}.$$

Indeed $a \equiv r_0 b \pmod{2M}$ and $b^{-1}M \equiv M \pmod{2M}$, while $Md \equiv M$ for odd d . A negative d can be oriented positive by negating v ; its height and child are unchanged.

Put $\tau = \min(a, b)$, $h = \max(a, b)$, $\sigma = a + b$ and $X = \max\{h, (M + 2ab)/\sigma\}$. Parameterize the real determinant line by $(-M/\sigma, M/\sigma) + t(a, b)$. If $M \leq h(h - \tau)$, then $X = h$. When $a = h$, the full interval $c \in [-h, h]$ gives $|d| = |(M + bc)/a| \leq h$; its length in t is 2. When $b = h$, use the full interval $d \in [-h, h]$ instead. If $M \geq h(h - \tau)$, then $X = (M + 2ab)/\sigma \geq h$. The points at $t = -2b/\sigma$ and $t = 2a/\sigma$ are

$$\left(-X, \frac{M - 2b^2}{\sigma}\right), \quad \left(\frac{-M + 2a^2}{\sigma}, X\right).$$

Both belong to $[-X, X]^2$: the required conditions are $M \geq b(b - a)$ and $M \geq a(a - b)$, which follow from the case assumption. The closed segment between them has parameter length 2. In either case it intersects the class $t_0 + 2\mathbb{Z}$. Thus the other child has height at most X .

Writing $x = \tau/h$, we obtain

$$hH \leq \max\left\{h^2, \frac{M + 2h^2x}{1 + x}\right\} \leq \max(M, 2h^2).$$

If $a = 0$, primitivity forces $b = 1$, the children are 0 and M , and their heights are 1 and M , so the same conclusion holds. Ties between parent minima need no special choice. $\square$

Define the exact finite quantities

$$\begin{aligned} C_M(t) &= \#\{r \bmod M : R_M(r) \leq t\}, \\ K_M &= \frac{1}{M} \sum_r [\ln(R_M(r)/s)]_+, \\ \mathcal{D}_M &= \frac{1}{M} \int_s^q C_M(t) \frac{dt}{t}, \\ \mathcal{L}_M &= \frac{1}{M} \sum_r [\ln(R_M(r)/q)]_+. \end{aligned} \quad (2.2)$$

We reserve D without a calligraphic letter for a dyadic content below. Taking logarithms in (2.1), and averaging over both children, gives

$$\mathbb{E}_{r \bmod 2M} \ln R_{2M}(r) \leq \frac{1}{2} \ln M + K_M. \quad (2.3)$$

There is no induction hypothesis on the parent mean.

For every $R > 0$,

$$[\ln(R/s)]_+ = \ln(q/s) - \int_s^q \mathbf{1}_{R \leq t} \frac{dt}{t} + [\ln(R/q)]_+.$$

Consequently

$$K_M = \frac{1}{2} \ln 2 - \mathcal{D}_M + \mathcal{L}_M. \quad (2.4)$$

It suffices to prove $K_M \leq \frac{1}{2} \ln 2$ at every parent length. For large lengths we prove $\mathcal{L}_M < \mathcal{D}_M$.

3 Short even vectors and the actual odd coset

For $r \bmod M$ let

$$\Lambda_r = \{(A, B) \in \mathbb{Z}^2 : A - rB \in M\mathbb{Z}\}.$$

Its basis $(M, 0), (r, 1)$ has determinant M .

Lemma 3.1. *If $R_M(r) > q$, there exist*

$$D = 2^j, \quad 1 \leq j \leq J, \quad b > 0 \text{ odd}, \quad \gcd(a, b) = 1, \quad D \max(|a|, b) \leq q$$

such that

$$a - rb = (M/D)\alpha_r, \quad \alpha_r \text{ odd}. \quad (3.1)$$

For each such triple, its cell $\mathcal{C}(D, a, b)$ of residues satisfying (3.1) has exactly $D/2$ elements. For each residue in that cell, $D(a, b)$ is primitive as an element of Λ_r .

Proof. Minkowski's convex body theorem [7] in the square of radius $q + \varepsilon$ gives a nonzero lattice vector of height at most $q + \varepsilon$. Discreteness permits passage to the closed radius q . Choose a shortest nonzero vector. Its denominator is nonzero, since a nonzero horizontal vector has height at least $M > q$. Orient it positive. The hypothesis $R_M(r) > q$ forces its denominator even, and its numerator is then even as well. A common odd factor could be divided out while preserving lattice membership, contradicting minimality. Thus its content is $D = 2^j \geq 2$; write it as $D(a, b)$ with $\gcd(a, b) = 1$. The congruence modulo $Q = M/D$, which is even, forces b odd. If $\alpha_r = (a - rb)/Q$ were even, division by 2 would again yield a shorter vector of Λ_r . This proves the asserted contact.

There is one solution of $a - rb \equiv 0 \pmod{Q}$. Its D lifts modulo M change α_r by $-bt$, $t \bmod D$. Exactly $D/2$ have odd contact. The coordinates of $D(a, b)$ in the displayed lattice basis are (α_r, Db) . Their gcd is 1: α_r is odd, and $\gcd(\alpha_r, b) = 1$ follows from $\gcd(a, b) = 1$ and $\gcd(Q, b) = 1$. This is lattice primitivity, despite the even integer content. $\square$

Lemma 3.2. *On a cell $\mathcal{C}(D, a, b)$ put $A = D|a|$, $B = Db$. Then*

$$R_M(r) \leq X := \frac{M + AB}{A + B}, \quad X \geq q. \quad (3.2)$$

On the axis $a = 0, b = 1$, the height is exactly $R_M(r) = M/D$.

Proof. For $a > 0$, lattice primitivity allows completion of $v = (A, B)$ to a basis with determinant M . Such completions $w = (c, d)$ satisfy

$$ad - bc = Q, \quad ad \equiv Q \pmod{b}, \quad \alpha_r d \equiv 1 \pmod{D}.$$

The last two congruences give one class of odd d modulo $B = Db$ by the Chinese remainder theorem. Invert a only modulo b ; no inversion modulo an even denominator is needed. For $b = 1$ the first congruence is empty. All completions differ by a multiple of v .

Put $x = M/(A + B)$. Let $d_- \leq x < d_+ = d_- + B$ be the adjacent denominators in this class, and write $\eta = x - d_- \in [0, B)$. On the determinant line, $c(d) = (Ad - M)/B$. The two heights are

$$\max(|c(d_-)|, |d_-|) = x + (A/B)\eta, \quad \max(|c(d_+)|, |d_+|) = x + B - \eta. \quad (3.3)$$

For clarity, the sign checks use $A, B \leq q$. If $d_- < 0$ and $A < B$, then $(B - A)|d_-| \leq (B - A)(B - x) \leq B^2 \leq M$, so $-c(d_-) \geq |d_-|$; the case $A \geq B$ is immediate. At $d_+ > x$, $-c(d_+) \leq d_+$. If $A > B$, then $c(d_+) \leq d_+$ follows from $(A - B)(x + B) \leq M$, equivalent to $A^2 - B^2 \leq 2M$. The other sign cases follow directly from $d_- \leq x < d_+$. The smaller expression in (3.3) is at most $x + AB/(A + B) = X$, since the two affine expressions meet at $\eta = B^2/(A + B)$. Negative odd denominators are oriented positive by negating both coordinates. Reflection handles $a < 0$. Finally $X/q \geq 1$ is equivalent to $(1 - A/q)(1 - B/q) \geq 0$.

If $a = 0$, primitivity gives $b = 1$ and $r = -Q\alpha_r$. Every marked numerator is a nonzero odd multiple of Q , so the height is at least Q . Take $c = -Q$ and a positive odd $d \leq D - 1$ representing $\alpha_r^{-1} \pmod{D}$. This has height Q , because $D \leq q \leq Q$. $\square$

Remark 3.3. Only coverage and the cell masses are needed for our upper bound. We do not require disjointness of the cells or replace their actual odd phases by continuous phase averages. The union bound below remains valid even without a uniqueness statement.

Let $V = \{r \neq 0 : v_2(r) \geq \lceil n/2 \rceil\}$. The axis cells are precisely these residues: for $D = 2^j$ their valuation is $k = n - j$ and their height is 2^k . This family is already present in Tian–Qi [1, Lemma 4]. Write $\mathcal{L}_V$ for its contribution to $\mathcal{L}_M$ and $\mathcal{L}_{\text{other}} = \mathcal{L}_M - \mathcal{L}_V$. For $0 < x, y \leq 1$ define

$$f(x, y) = \ln \frac{1 + xy}{x + y}, \quad P_f(T) = \sum_{\substack{1 \leq a, b \leq T \\ b \text{ odd}, \gcd(a, b) = 1}} f(a/T, b/T).$$

The kernel is nonnegative. Every non-axis residue with height above q is covered by Lemma 3.1, and Lemma 3.2 bounds its energy by $f(D|a|/q, Db/q)$. Each cell has $D/2$ residues and both signs of a give the weight D . Therefore, using an upper bound by a union of cells,

$$\mathcal{L}_{\text{other}} \leq U_M := \frac{1}{M} \sum_{j=1}^J 2^j P_f(q/2^j). \quad (3.4)$$

Including cells whose height is at most q only enlarges this upper bound; all summands are nonnegative.

4 An explicit estimate of the lattice sum

Let

$$I = \int_{[0,1]^2} f(x, y) dx dy, \quad F(T) = \sum_{\substack{1 \leq a, b \leq T \\ b \text{ odd}}} f(a/T, b/T).$$

The integral is finite: $0 \leq f(x, y) \leq -\ln \max(x, y)$, so $I \leq 1/2$.

Lemma 4.1. *For every real $T > 0$,*

$$\frac{I}{2}T^2 - T \leq F(T) \leq \frac{I}{2}T^2, \quad I = \frac{\zeta(2) - 1}{2} = \frac{\pi^2}{12} - \frac{1}{2}. \quad (4.1)$$

Proof. For fixed $0 < x \leq 1$, extend $f(x, y)$ to $y \geq 0$. Its derivatives satisfy

$$\partial_y f = \frac{x^2 - 1}{(1 + xy)(x + y)} \leq 0, \quad \partial_y^2 f = \frac{1}{(x + y)^2} - \frac{x^2}{(1 + xy)^2} \geq 0.$$

The last assertion follows from $1 + xy \geq x(x + y)$, also beyond $y = 1$. Odd $b \leq T$ are the midpoints of intervals $[(b - 1)/T, (b + 1)/T]$. Convexity gives

$$\frac{2}{T} \sum_{b \leq T, b \text{ odd}} f(x, b/T) \leq \int_0^{Y_T} f(x, y) dy, \quad Y_T = \frac{2\lceil [T]/2 \rceil}{T}.$$

If $Y_T \leq 1$, the omitted portion is nonnegative. If $Y_T > 1$, the added portion is nonpositive, because $f(x, y) \leq 0$ for $y > 1$. Thus this integral is at most $\Phi(x) = \int_0^1 f(x, y) dy$. The function Φ is decreasing and nonnegative, so its right Riemann sum at a/T , $1 \leq a \leq [T]$, is at most TI . This proves the upper bound for real T , including the last odd cell.

For $T \geq 1$, the clipped rectangles $[a/T, (a + 1)/T] \times [b/T, (b + 2)/T]$ with the same indices cover $[1/T, 1]^2$ with disjoint interiors. Monotonicity bounds the integral there by $2F(T)/T^2$. The omitted vertical and horizontal strips have total integral at most $2/T$, using $f \leq -\ln y$ and $f \leq -\ln x$, respectively. This proves the lower bound. For $0 < T < 1$, $F(T) = 0$ and the lower bound is nonpositive.

For $z \geq 1$, the area of the region $(1 + xy)/(x + y) \geq z$ is

$$\mathcal{A}(z) = \int_0^{1/z} \frac{1 - zx}{z - x} dx = 1 - (z^2 - 1) \ln \frac{z^2}{z^2 - 1} = \sum_{k \geq 1} \frac{z^{-2k}}{k(k + 1)}.$$

The endpoint $z = 1$ is interpreted by a limit. Layer cake and Tonelli's theorem give

$$I = \int_1^\infty \mathcal{A}(z) \frac{dz}{z} = \frac{1}{2} \sum_{k \geq 1} \frac{1}{k^2(k + 1)} = \frac{\zeta(2) - 1}{2}.$$

The last expression uses $1/(k^2(k + 1)) = 1/k^2 - 1/k + 1/(k + 1)$ and Euler's classical evaluation $\zeta(2) = \pi^2/6$. $\square$

A primitive pair cannot have either 3 or 5 as a common divisor. Since $f \geq 0$, inclusion-exclusion in the larger set which excludes only those two common divisors gives

$$P_f(T) \leq F(T) - F(T/3) - F(T/5) + F(T/15).$$

Dividing by an odd common divisor preserves the odd denominator. Using the upper bound in (4.1) in the positive terms and the lower bound in the negative terms yields

$$P_f(T) \leq \frac{32I}{75}T^2 + \frac{8}{15}T.$$

This includes scaled radii smaller than 1. Substitution in (3.4) gives the uniform finite estimate

$$\mathcal{L}_{\text{other}} \leq \frac{32I}{75}(1 - 2^{-J}) + \frac{8J}{15q}. \quad (4.2)$$

5 Central credit and the infinite range

For integer $T \geq 1$, define the signed primitive count, with the axis once,

$$A(T) = \#\{(a, b) : |a| \leq T, 1 \leq b \leq T, b \text{ odd}, \gcd(a, b) = 1\}.$$

Lemma 5.1. *For every integer $T \geq 1$,*

$$A(T) \geq \frac{8}{\pi^2} T^2 - \frac{3}{4} T \ln T - \frac{11}{4} T - \frac{1}{4}. \quad (5.1)$$

For $n \geq 2$, $T_0 = \lfloor q/\sqrt{2} \rfloor$ satisfies $C_M(T_0) = A(T_0)$, and

$$\mathcal{D}_M \geq \frac{2 \ln 2}{\pi^2} - \frac{4\sqrt{2} \ln 2 / \pi^2 + (\ln 2 / (2\sqrt{2}))((3/4) \ln q + 11/4)}{q} - \frac{\ln 2}{8q^2}. \quad (5.2)$$

Proof. Möbius inversion over odd common divisors gives the exact formula

$$A(T) = \sum_{\substack{g \leq T \\ g \text{ odd}}} \mu(g) (2 \lfloor T/g \rfloor + 1) \left\lceil \frac{\lfloor T/g \rfloor}{2} \right\rceil. \quad (5.3)$$

For $l = \lfloor x \rfloor$ the raw count $B(l) = (2l+1) \lfloor l/2 \rfloor$ equals $l^2 + l/2$ when l is even and $l^2 + 3l/2 + 1/2$ when it is odd. Hence $|B(l) - x^2| \leq 3x/2 + 1/2$. Also

$$\sum_{\substack{g \leq T \\ g \text{ odd}}} \frac{1}{g} \leq 1 + \frac{1}{2} \ln T, \quad \#\{g \leq T : g \text{ odd}\} \leq (T+1)/2.$$

For the first inequality, $1/(2k+1) \leq \frac{1}{2} \ln((2k+1)/(2k-1))$ for $k \geq 1$ telescopes. The raw error in (5.3) is at most $(3/4)T \ln T + (7/4)T + 1/4$. The absolutely convergent Euler product gives $\sum_{g \text{ odd}} \mu(g)/g^2 = 8/\pi^2$; truncating it at T costs at most $T^2 \sum_{g > T} g^{-2} \leq T$. This proves (5.1).

Two distinct primitive representations in the T_0 box of one residue have a nonzero determinant divisible by M , unless they coincide. The determinant has absolute value at most $2T_0 B_0$, where B_0 is the largest odd integer at most T_0 . If $2T_0^2 < M$, this is less than M . If $2T_0^2 = M$ and $n \geq 2$, T_0 is a power of 2 and is even, so $B_0 = T_0 - 1$ and the strict inequality still holds. A zero determinant gives the same primitive fraction with positive denominator. Thus the residue map is injective; every covered residue has a primitive representation in the box, so $C_M(T_0) = A(T_0)$. The excluded case $n = 1$ has the collision $(1, 1), (-1, 1)$.

By monotonicity $\mathcal{D}_M \geq (\ln 2)/(2M) A(T_0)$. Use $T_0^2 \geq q^2/2 - \sqrt{2}q$, $T_0 \ln T_0 \leq (q/\sqrt{2}) \ln q$ and $T_0 \leq q/\sqrt{2}$ in (5.1). This yields (5.2). $\square$

The axis mass at valuation k is 2^{-k-1} . Consequently

$$\frac{\mathcal{L}_V}{\ln 2} = \sum_{k=\lceil n/2 \rceil}^{n-1} 2^{-k-1} (k - n/2) = \begin{cases} 2^{-\ell} - (\ell+1)2^{-2\ell}, & n = 2\ell, \ell \geq 1, \\ 3 \cdot 2^{-\ell-2} - (2\ell+3)2^{-2\ell-2}, & n = 2\ell+1, \ell \geq 0. \end{cases} \quad (5.4)$$

In particular $\mathcal{L}_V \leq \sqrt{2} \ln 2 / q$ for all $n \geq 1$. These are finite geometric sums, including the zero energies at $n = 1, 2$.

Proposition 5.2. *For every parent length $n \geq 26$, $\mathcal{L}_M < \mathcal{D}_M$, hence $K_M < \frac{1}{2} \ln 2$.*

Proof. Put

$$c = \frac{32I}{75} = \frac{16}{75} \left(\frac{\pi^2}{6} - 1 \right), \quad \delta = \frac{2 \ln 2}{\pi^2} - c.$$

Equations (4.2), (5.2), (5.4), discarding the favorable term $-c2^{-J}$, give $\mathcal{D}_M - \mathcal{L}_M \geq \delta - E(q, J)$, where

$$E(q, J) = \frac{8J/15 + 4\sqrt{2} \ln 2 / \pi^2 + (\ln 2 / (2\sqrt{2}))((3/4) \ln q + 11/4) + \sqrt{2} \ln 2}{q} + \frac{\ln 2}{8q^2}.$$

Use $3 < \pi < 22/7$, $1 < \sqrt{2} < 3/2$, and $\ln 2 > 2/3 + 2/81 = 56/81$ from the positive atanh series. Then

$$\delta > \frac{1372}{9801} - \frac{304}{2205} = \frac{45756}{21611205} > \frac{1}{500}.$$

Since $q \geq 2^J$ and $\ln q < J + 1$, the same elementary bounds give

$$E(q, J) \leq \left(\frac{109}{120} J + \frac{47}{12} \right) 2^{-J} + \frac{1}{8} 2^{-2J}.$$

At $J = 13$ the right side is $629/327680 + 1/536870912 < 1/520 + 1/13000 = 1/500$. Both terms decrease for $J \geq 13$; for the first, $aJ + b > a$ with $a = 109/120$, $b = 47/12$. Thus $n \geq 26$ implies $E < 1/500 < \delta$. $\square$

6 Finite certificates and completion of the proof

The remaining premise is $K_{2^n} \leq \frac{1}{2} \ln 2$ for $1 \leq n \leq 25$. There are two different finite interfaces, described here so the computation has an exact mathematical meaning. No decision uses floating logarithms.

For $n = 1, \dots, 12$, certificates give the height h_r and a primitive witness (a_r, b_r) for every residue. The checker verifies $b_r > 0$ odd, $a_r \equiv rb_r$, and $\max(|a_r|, b_r) = h_r$. To prove minimality, for every positive odd $b < h_r$ it computes the least absolute representative of $rb \bmod M$ and checks that it is at least h_r . This excludes every improving pair. Let $c_n = \#\{r : 2h_r^2 > M\}$ and $P_n = \prod_{2h_r^2 > M} h_r^2$. Then the required comparison is exactly

$$P_n \leq 2^{M+(n-1)c_n}. \quad (6.1)$$

This follows by exponentiating $2MK_M \leq M \ln 2$. The full domain has $\sum_{n=1}^{12} 2^n = 8190$ minima. For diagnostic support, the checker also verifies $\prod_r h_r^2 \leq M^M$, but (6.1) is the premise used with the child lemma.

For $n = 13, \dots, 25$, set $T_0 = \lfloor \sqrt{M/2} \rfloor$ and compute $A = A(T_0)$ exactly by (5.3). Put

$$V_2 = \sum_{\substack{0 \leq k < n \\ 2k > n}} 2^{n-k-1} (2k - n) = \frac{2M\mathcal{L}_V}{\ln 2}, \quad x_{D,a,b} = \frac{(M + D^2 ab)^2}{MD^2(a+b)^2} \geq 1.$$

Thus $2MU_M = \sum_{D,a,b} D \ln x_{D,a,b}$, with $D = 2^j$, $1 \leq j \leq J$, $1 \leq a, b \leq q/D$, b odd, $\gcd(a, b) = 1$. Using $S = 2^{40}$, let $\ell^+(x) \geq S \ln x$ and $\ell_2^- \leq S \ln 2$ be the integer bounds of Appendix A. The sufficient certificate inequality is

$$\sum_{D,a,b} D \ell^+(x_{D,a,b}) \leq (A - V_2) \ell_2^-, \quad A > V_2. \quad (6.2)$$

Indeed (6.2) implies $U_M + \mathcal{L}_V \leq A \ln 2 / (2M) \leq \mathcal{D}_M$. The integer margin, right side minus left side, is listed below. It is a scaled certificate margin, not the exact real difference $\mathcal{D}_M - \mathcal{L}_M$.

Parent n	Integer margin	Parent n	Integer margin
13	200913119387100	20	22717214856196764
14	343712368064520	21	45370620212212808
15	778422546045060	22	90744127844519936
16	1668697844541872	23	181329491713437076
17	2824453720758008	24	359749179298450700
18	5916038348384600	25	720718942077388912
19	11572174724235024		

The fixed panel contains 9 068 656 positive primitive pairs, out of 11 182 008 raw pairs. The input and replay map in the companion separate these essential certificates from later diagnostic controls.

Proof of Theorem 1.1. The small certificates prove the sufficient bound on K_M for parents $n = 1, \dots, 12$; the directed-log certificates prove it for $n = 13, \dots, 25$. Proposition 5.2 proves it for $n \geq 26$. Equation (2.3) therefore gives $\mathbb{E} \ln R_N \leq (N/2) \ln 2$ for every $N \geq 2$, using parent length $n = N - 1$. For $N = 1$, both residues have height 1. Divide by $\ln 2$. $\square$

7 Scope, reproducibility and remaining questions

The theorem concerns the ordinary mean over all finite binary strings. It gives no upper bound for every prescribed infinite 2-adic number and no statement about all Collatz trajectories. The variance part of [4, Question 3.1], sharper constants, and extensions to other bases are not proved here.

The structural ingredient is the compensation between the two dyadic lifts, followed by an upper bound in the actual odd completion class. Bézout, Minkowski, Möbius inversion, Euler products, convexity and Tonelli are classical tools. The short count and high-valuation family have explicit antecedents credited above. The contribution is the exact mean bound and the estimates which close its sign. Priority remains a separate source question.

The companion consists of this standalone source, a rendered PDF, `companion/verify_mean.py`, `companion/finite_certificates.json`, a claim map and reproduction instructions. From the repository root run

```
python -B scripts/reproduce.py
python -B -O scripts/reproduce.py
```

The two runs must produce the same deterministic mathematical JSON. The checker verifies level coverage, witnesses and minima, exact products, directed-log sums, populations and margins, with explicit failures that remain active under Python optimization. It checks only the finite premises; reading the lemmas above is necessary to assess the all-length proof. The source compiles independently of the research reports and bibliography databases.

AI assistance. OpenAI GPT models assisted mathematical exploration, proof criticism, source comparison, editing and reproducibility checks. The AI review passes shared exposure to the research material; they do not constitute independent human peer review or formal verification. The author retains responsibility for the arguments and attributions.

A Directed logarithms using integer arithmetic

We specify the rounding used in (6.2). For a rational $v = u/w$ with $1 \leq v \leq 2$ let $z = (u - w)/(u + w) \in [0, 1/3]$. Then

$$\ln v = 2 \sum_{j=0}^{11} \frac{z^{2j+1}}{2j+1} + 2 \sum_{j=12}^{\infty} \frac{z^{2j+1}}{2j+1}.$$

To compute an upper, replace Sz by its ceiling, compute its square divided by S with upward rounding, start the scaled power with that ceiling, and at each step round both division by $2j+1$ and the multiplication by the scaled square divided by S upwards. Twice the twelve-term sum is an upper for the partial sum. The exact tail, using $z \leq 1/2$, is bounded by

$$\frac{2z^{25}}{25(1-z^2)} \leq \frac{8}{3 \cdot 25 \cdot 2^{25}}.$$

Add the integer ceiling of $8S/(3 \cdot 25 \cdot 2^{25})$. For a lower, use downward rounding at every step and omit the nonnegative remainder. For $v = 1$, return exactly zero. These rules give a lower ℓ_2^- and upper ℓ_2^+ for $S \ln 2$.

For an arbitrary rational $x = u/w \geq 1$, find the unique integer $k \geq 0$ with $2^k w \leq u < 2^{k+1} w$, by integer bit lengths and a comparison. Apply the unit-interval upper to $u/(2^k w)$ and add $k\ell_2^+$. Every operation is an integer comparison, product, floor or ceiling. Since all rounding maps are monotone on nonnegative inputs, the computed scaled powers bound the true scaled powers by induction. The uniform tail bounds the true analytic remainder; it does not assume that an upward-rounded approximation of z still equals z . This proves the directed inequalities used by every summand of the finite bridge.

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